Machine Learning II
Text Classification

POLI3148 Data Science in Politics and Public Administration

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Review: What is Machine Learning?

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions.

Advances in the field of deep learning have allowed neural networks to surpass many previous approaches in performance.

Source: Wikipedia

Review: Classification Model

Definition: A model whose prediction is **a class**. For example, the following are all classification models:

- A model that predicts an input sentence's language (French? Spanish? Italian?).
- A model that predicts tree species (Maple? Oak? Baobab?).
- A model that predicts the positive or negative class for a particular medical condition.
- *A model that predicts if an official can get promotion*
- *A model that predicts if a war will happen*
- *A model that predicts which candidate can win the election*

In contrast, ***regression models*** predict numbers rather than classes.

Source: Google Machine Learning

Text Classification Model

Definition: A ML model whose **input (X)** is text and predicted **target (y)** is **a class**. For example, the following are all classification models:

A model predicting…

- if a social media post contains hate speech
- if a official document discusses political violence
- if a question in a preference conference contains positive/ negative/ neutral sentiment
- if a question to ChatGPT is inappropriate, for which ChatGPT should refuse to answer

Workflow

Data Input:

- Load data
- Define **X** and **y**
- Handle missing values
- Split data into training and test sets
- Re-formats for Machine Learning tool

Data Processing:

- Define model
- Train model

Data Output:

- Evaluate model
- Interpret model
- Discussion: What your results say about the substantive topics

How Text Classification is special:

Define **X** and **Y**
Interpret model

Running Example: Understanding Chinese Diplomacy w/ Text Analysis

Chinese Ministry of Foreign Affairs Press Conferences Corpus

- Data: https://dataverse.harvard.edu/dataset.x ... i:10.7910/DVN/BAKGET
- Paper: https://link.springer.com/article/10.1 ... 3

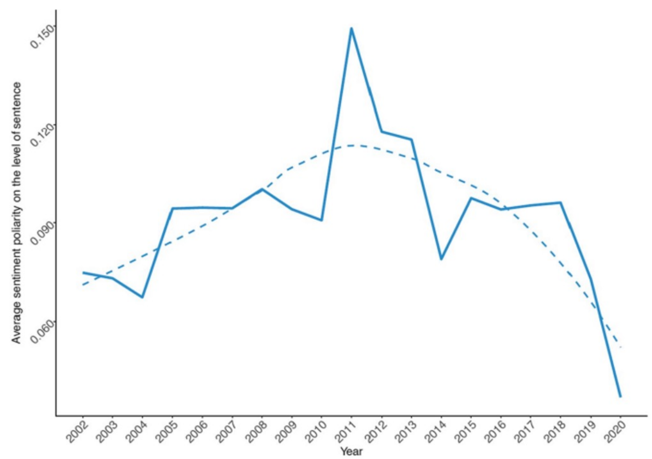

Fig. 4 Change in sentiment polarity towards the US in Chinese foreign policy discourse. Note: Solid line visualizes raw sentiment averages on the level of sentences. Dashed line visualizes smoothed trend (using loess function [Local Polynomial Regression Fitting])

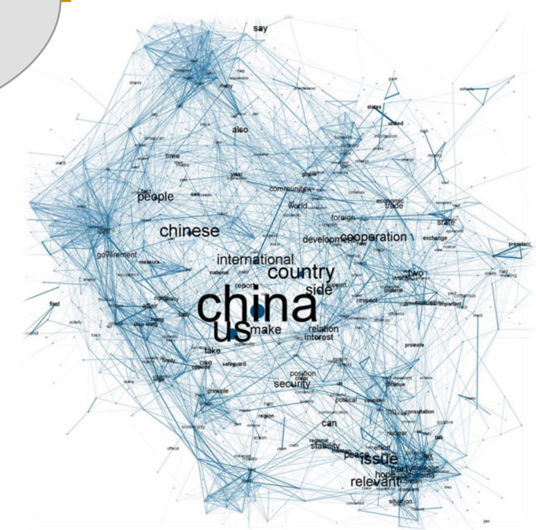

Fig. 6 Semantic network tracing US-China discourse under President Xi Jinping

What Have We Done with These Data?

- Extracting Textual Features
    - Linguistic features
        - Tokenization
        - Lemmatization
        - Stopword removal
    - Named Entities
- Topic Modeling

We used a bag-of-word approach to extract textual features and conducted some macro-level (corpus-level) analysis

We have not done any **micro-level** analysis

Text Classification Application: Sentiment Analysis

Task:

Develop a ML classifier to identify "aggressive" questions in the MoFA Q&A dataset

Steps:

- Manually label a small portion of the MoFA Q&A
- Train and evaluate a ML classifier
- Apply the ML classifier to Q&A that are not manually labeled

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Get y

Get **y** from the raw data

- Create a “**q_aggressive**” column using the existing “**q_sentiment”** column
    - Normally, y is obtained through manual labeling
    - We take advantage of the dataset, which obtain the sentiment of questions using other machine learning methods
    - For this task, we define it as questions whose sentiment scores are at the bottom 25 percentile

Get X

Get **X** from the linguistic feature: Load the linguistic features we previously extracted from questions using *spaCy*

- Remove "useless" tokens (stopwords, punctuations, numbers, etc.)
- Reshape the table to make one row per document (a *document* is a question in our case)
- Create a **document-term matrix** using CountVectorizer

Remember this?
See Lecture 8

	she	loves	pizza	is	delicious	a	good	person	people	are	the	best
She loves pizza, pizza is delicious	1	1	2	1	1	0	0	0	0	0	0	0
She is a good person	1	0	0	1	0	1	1	1	0	0	0	0
good people are the best	0	0	0	0	0	0	1	0	1	1	1	1

Follow the code.

See how to obtain y, X, X_names

The rest are almost the same

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Data Processing:

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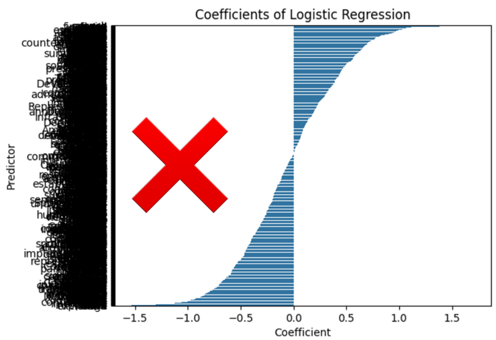
Data Output:

- Evaluate model
- **Interpret model**
- Discussion: What your results say about the substantive topics

Interpret Model: Be Selective

- With text classification, or other classification that contains a lot of predictors, you cannot interpret ALL predictors (thousands of them)
- What to do
  - Interpret only a subsample of predictors
  - Sort the coefficients/ feature importance
  - For logistic regression: Interpret those with largest coefficients
  - For Decision Trees/ Random Forest: Interpret those with highest feature importance



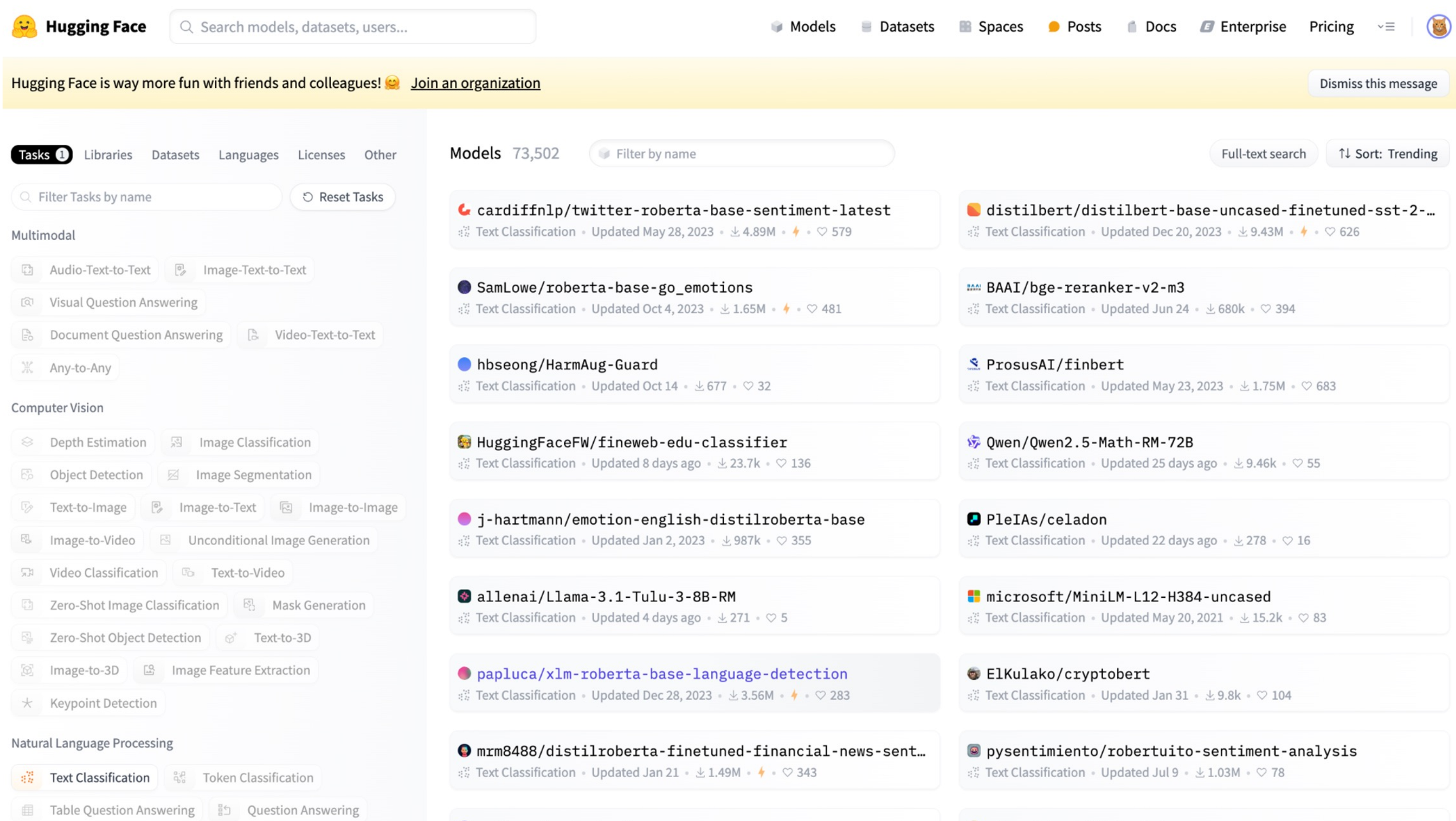
Try Decision Trees and Random Forest yourself

Finally,

A Postmodern Approach to Machine Learning

Remember Hugging Face?

Source: https://huggingface.co/models?pipeline_tag=text-classification&sort=trending



Using Resources from Hugging Face

- A playground of ML models… including text classification models
- Pros
  - State-of-the-art: Latest published works in ML and AI publish their models here
  - User-friendly
- Cons
  - Not all are reliable, need to be selective (if unsure, use the most popular ones)
  - Some are black boxes (in terms of model architecture and training data)
  - Authors can delete the resources (don’t assume it will always be available.)

Let's try a simple text classification task together